

AlgebraOfGraphics.jl: An algebraic grammar of graphics for Julia, powered by Makie.jl

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Software

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Summary

AlgebraOfGraphics.jl is a Julia package for declarative data visualization built on top of the Makie.jl plotting ecosystem (Danisch & Krumbiegel, 2021). It provides a high-level grammar in which datasets, variable mappings, analyses, and visual specifications are represented as first-class objects that can be combined with the algebraic operators `+` and `*`. The distributive property of these operators makes it possible to factor out information that is shared across layers of a figure, eliminating much of the repetition that users encounter when composing layered plots in lower-level plotting APIs. A plot specification is rendered into a full Makie figure by a single draw call, and because the result is an ordinary Makie figure, users retain full programmatic access to every axis, legend, and plot object for further customization.

Statement of need

Exploratory statistical visualization requires expressing, in as few lines as possible, the relationship between variables in a dataset and the visual attributes of a figure. Users typically want to map categorical variables to colors or markers, split a figure into small multiples, overlay raw data with summary statistics, attach legends and colorbars, and label axes meaningfully — all while iterating rapidly on the underlying question. Doing this directly with a general-purpose plotting library such as Makie.jl is technically possible but cumbersome: the user is forced to manage groupings, allocate palette entries, compute limits across facets, and assemble legends manually, and the resulting code obscures the analytic question behind a large amount of bookkeeping.

R's ggplot2 (Wickham, 2016) popularized the grammar of graphics (Wilkinson, 2005) as a solution to this problem, and ports and successors exist in many languages. In Julia, however, there was no well-maintained grammar-of-graphics front-end for Makie.jl, which has become one of the most capable native Julia plotting libraries, with support for 2D and 3D graphics, interactivity, and multiple backends (Danisch & Krumbiegel, 2021). AlgebraOfGraphics.jl fills this gap, and does so with two design choices that distinguish it from existing grammar-of-graphics libraries.

First, AlgebraOfGraphics.jl represents plot specifications as elements of an algebraic structure. Users combine building blocks — `data(table)`, `mapping(:x, :y, color = :group)`, `visual(Scatter)`, analyses such as `linear()` or `density()` — with `*` (which merges specifications) and `+` (which superimposes layers). Because `*` distributes over `+`, information that is shared across layers can be written once and factored out. For example,

```
data(penguins) * mapping(:bill_length_mm, :bill_depth_mm, color = :species) *  
  (visual(Scatter) + linear())
```

expresses a scatter plot with a linear fit per species, with the dataset, mapping, and color grouping factored out of the superimposed layers. This algebraic factoring is rarely possible in

imperative grammar-of-graphics libraries, where each layer typically re-specifies its own data and aesthetics. **Figure 1** illustrates how a plot is built incrementally and how the individual building blocks combine into a single expression.

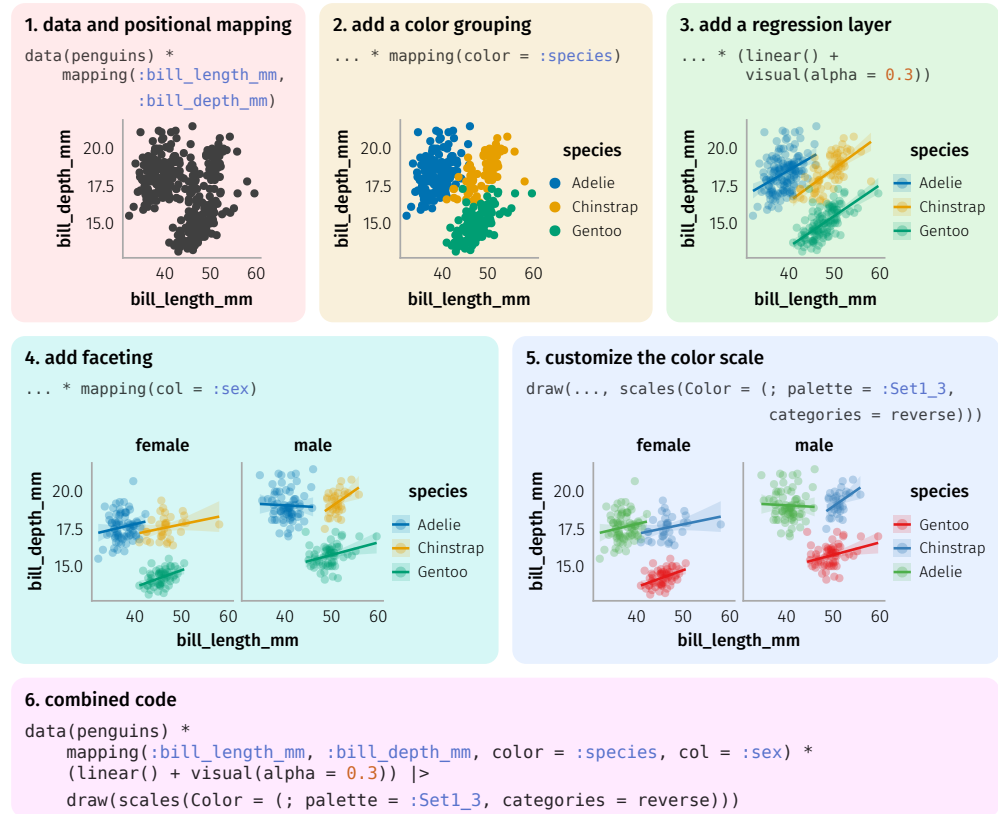


Figure 1: Building a plot incrementally with * and +. Each panel shows a step in the construction of a visualization of the Palmer penguins dataset: starting from data and a positional mapping, we add a color grouping, a regression layer, a facet by sex, and finally a customized color scale with reversed category order. The bottom panel shows the equivalent expression as a single pipeline.

Second, AlgebraOfGraphics.jl is a thin grammar layer over a native Julia plotting library rather than a self-contained rendering engine. The result of draw is a Makie figure, which means users can freely mix AlgebraOfGraphics layers with hand-authored Makie plots, add or modify axes after the fact, and leverage Makie's interactivity and multiple backends without leaving Julia. This matters in practice because published figures are rarely identical to their first draft: authors routinely tweak labels, add annotations, adjust legends, and rearrange panels. Doing this programmatically keeps the figure fully reproducible, in contrast to workflows that finish a figure in Illustrator or Inkscape.

State of the field

Several existing tools occupy adjacent parts of the space:

- ggplot2 (Wickham, 2016) is the canonical grammar-of-graphics implementation. Its underlying representation is an R grob tree that is comparatively opaque once rendered, making it difficult to adjust properties of the figure which are not directly exposed by ggplot's interface.

- `plotnine` (Kibirige, 2024) and `seaborn` (Waskom, 2021) occupy similar ground in Python. They can be called from Julia via `interop`, but cannot be extended from Julia, so native Julia data types and plotting recipes do not integrate with them. Tweaking the resulting figures further requires familiarity with `matplotlib`, which adds another layer of tooling that Julia users may not be familiar with.
- `Gadfly.jl` (Jones et al., 2014) was an earlier grammar-of-graphics library for Julia with an elegant design, but its output options and interactivity story remained limited and its development has largely stalled.
- `VegaLite.jl` (Anthoff & VegaLite.jl contributors, 2021) wraps the Vega-Lite declarative specification. The generated artefact is a JSON spec, which means figures are portable across renderers but cannot be freely manipulated with native Julia plotting code.
- `TidierPlots.jl` (Boyes, 2024) is a recent effort to port `ggplot2`'s syntax onto `Makie.jl`. It shares `AlgebraOfGraphics.jl`'s Makie substrate and overlaps substantially in the set of plots that can be produced. The key difference is `AlgebraOfGraphics.jl`'s algebraic model: `+` and `*` combined with distributivity offer a compositional style that differs from `ggplot2`'s imperative `+`-chains.
- `StatsPlots.jl` extends the recipe-based `Plots.jl` ecosystem with statistical plots, but it is not a grammar-of-graphics library and does not separate data, mapping, and visual specification.

Software design

A plot specification is represented internally as a `Layer` or sum of layers. Each layer carries a dataset, a mapping from column references to aesthetic roles, a visual specification (which Makie recipe to use and with which fixed attributes), and optionally an analysis that transforms the data before plotting (e.g. kernel density estimation, a linear fit, histogramming). During draw, layers are grouped, analyses are applied, and the resulting data are handed to a scales system that resolves categorical and continuous variables into concrete Makie attributes across all axes, legends, and colorbars of the figure.

The scales system is a central piece of the current architecture and is the result of a major refactor of the original codebase. In the first versions of `AlgebraOfGraphics.jl` the intent was to be a thin dispatcher that forwarded positional and keyword arguments almost directly to Makie recipes. Experience showed that keeping an implicit interface between `AlgebraOfGraphics` and Makie plotting functions was not sustainable, not all functions subscribe to the pattern that `X` and `Y` are the first two positional arguments, not all functions have only a single color attribute called `color`. Introducing an explicit scales system resolved a large class of latent bugs and unlocked features that are otherwise difficult to express, including multiple independent color scales with their own colorbars within a single figure, multiple independent `X` or `Y` scales across neighboring facets, and correct label and tick management for empty or partially populated facets.

Because the connection between a Makie recipe and the AoG scale machinery is established through multiple dispatch on the recipe type, third-party Makie plotting recipes can register themselves with `AlgebraOfGraphics.jl` from outside the package, making the system fully extensible: any package that defines a custom Makie recipe can opt it into the grammar without modifications to `AlgebraOfGraphics.jl`. In combination with Julia's package extension mechanism, available since Julia 1.9, this support can be added conditionally — a downstream package can ship AoG integration that activates only when both `AlgebraOfGraphics` and the recipe-providing package are loaded, without taking on a hard dependency on either side.

Other notable design choices include: support for multiple tabular formats via the `Tables.jl` interface (Quinn & Tables.jl contributors, 2021) as well as “pre-grouped” collections of arrays for data that is not naturally tabular; wide-data mappings via `dims`, avoiding the need to reshape data before plotting; element-wise column transformations inside mapping, which interact correctly with grouping and faceting; a flexible pagination system for plots that would

107 otherwise contain too many facets to display at once; and first-class support for units through
108 Unitful.jl and DynamicQuantities.jl.

109 Research impact

110 AlgebraOfGraphics.jl is used across several communities. In pharmacometrics, it is the
111 recommended visualization tool in the Pumas software stack (Rackauckas et al., 2020). Other
112 software like UnfoldMakie.jl (Mikheev & Ehinger, 2025), a toolbox for visualization of event-
113 related EEG analyses, has been built on top of it. It is covered in open educational resources
114 including the *Julia Data Science* book (Storopoli et al., 2021) and John Verzani's *Using Julia*
115 *for Introductory Statistics*. The package has accumulated over 500 stars on GitHub and is
116 regularly discussed on the Julia Discourse. Because no canonical citation has existed until now,
117 adoption in the scientific literature has been under-reported; this paper is intended to provide
118 a stable reference for use in future publications.

119 AI usage disclosure

120 The manuscript was drafted with the assistance of a large language model (Anthropic's Claude)
121 acting as a writing collaborator. All claims, structure, and wording were reviewed and edited by
122 the authors. Within the software itself, no AI-generated code was merged during the first five
123 years of development; AI coding assistance has been used only for a small fraction of recent
124 contributions and is always reviewed by a human maintainer before merging.

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